

SET 10

1. Configure a hub-based network and explain the differences in traffic handling compared to a switch-based network.
2. Using Wireshark, Capture and analyze the Packets that uses UDP for different header fields and payload of the concerned packet.
3. Design a network for a fictional company with different departments, servers, and security features using Packet Tracer.
4. Create an IoT-based smart gardening system using Packet Tracer, which monitors soil moisture, temperature, and controls water flow.

1) Aim:-

To configure a simple hub-based network and understand how traffic is compared with a Switch-based Network

Topology

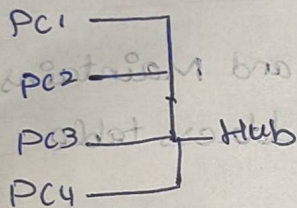
Use:

★ 1 Hub

★ 4 PCs

★ Ethernet Cables

Example:



IP addressing

Device	IP Address	Subnet Mask
PC1	192.168.1.1	255.255.255.0
PC2	192.168.1.2	255.255.255.0
PC3	192.168.1.3	255.255.255.0
PC4	192.168.1.4	255.255.255.0

Testing:

From PC1, open command prompt and run:

Ping 192.168.1.2

Hub	Switch
Works Mainly at Physical Layer (Layer 1)	Works mainly at Data Link Layer (Layer 2)
Send incoming data to all Ports	Sends frames only toward ^{Destination} address table Ports
Does not learn MAC addresses	Learns and Maintains MAC address table
Low efficiency	High efficiency

Conclusion:

A hub broadcasts incoming traffic to all connected devices, whereas a switch intelligently forwards ethernet frame based on MAC addresses

2) Aim:

To capture UDP packets and analyze their header fields and payload

Procedure:

1. Install and open Wireshark
2. Select the active network interface, such as WiFi
3. Start packet capture
4. Generate UDP traffic. For example, DNS queries
5. Stop the capture after packets appear
6. Apply the display filter

Field	Description
Source port	Port No of sending application
Destination port	Port No of receiving application
Length	Total length UDP header + data
Checksum	Used for error detection
Payload	Actual application data carried by UDP

The UDP header is 8 bytes long

Wireshark Analysis

Select a UDP packet and expand:

Use Datagram protocol

- ★ Source port
- ★ Destination port
- ★ Length
- ★ Checksum
- ★ UDP payload / data

Conclusion:-

Wireshark allows us to capture packets and examine their header fields and payload. UDP is connectionless and does not provide reliability mechanisms of TCP.

3) Aim:-

To design a secure network for a fictional company containing different department servers.

Company:-

Assume the company has three departments:

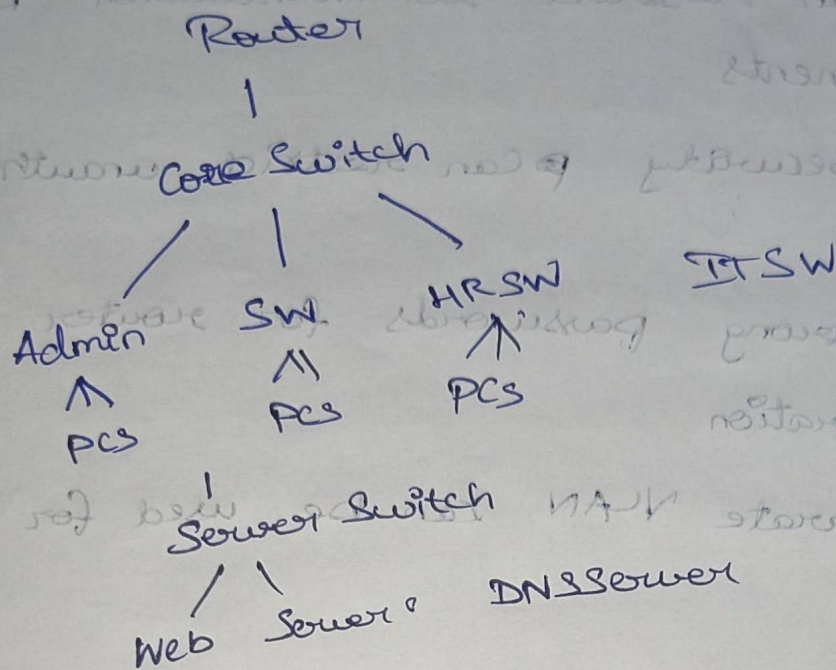
- ★ Administration
- ★ HR
- ★ IT

Device Required:-

- ★ 1 Router
- ★ 3 Switches
- ★ PCs for each department
- ★ 2 Servers

Ethernet Cable

Suggested Topology:-



IP addressing:-

Department	Networks
Administration	192.168.10.0/24
HR	192.168.20.0/24
IT	192.168.30.0/24
Servers	192.168.40.0/24

Examples

Admin PC1 → 192.168.10.10

HR PC1 → 192.168.20.10

IT PC1 → 192.168.30.10

Servers PC1 → 192.168.40.10

DNS Server → 192.168.40.20

Security features:-

- 1) VLANs can separate departments
- 2) ACLs can restrict communication between departments
- 3) Port Security can restrict unauthorized devices
- 4) Use strong passwords for router and switch administration
- 5) A separate VLAN can be used for servers

Conclusion:

The proposed network separates departments using VLANs and subnets while providing centralized server access and security through ACLs and port security.

01.01.801.001 < 104 0310A

01.02.801.001 < 104 914

01.03.801.001 < 104 714

01.04.801.001 < 104 0002

01.05.801.001 < 104 0002

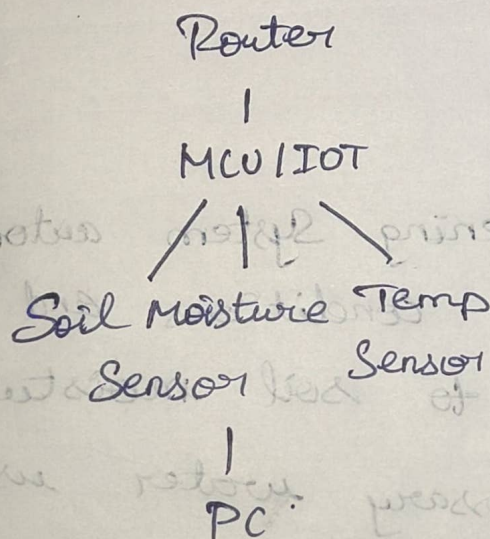
47 Aim :-

To create an IOT smart gardening system that monitors soil moisture and temperature and automatically controls water flow

Components :-

- ★ IOT MCU device
- ★ Soil Moisture Sensor
- ★ Temperature Sensor
- ★ Water Sprinkler
- ★ Home Gateway
- ★ PC
- ★ Network Switch

Basic Topology :-



Working:-

- ★ The soil moisture sensor measures the moisture level
- ★ The temperature sensor measures surrounding temperature
- ★ The IoT controller receives sensor readings
- ★ If soil moisture falls below a specified threshold, the controller turns ON water Pump
- ★ When sufficient moisture is detected, the Pump is turned OFF
- ★ Sensor Values can be monitored through Pc

Conclusion:-

The IoT gardening System automatically monitors environmental conditions and controls irrigation according to soil moisture. This reduces unnecessary water usage and allows remote monitoring through the network.